Kripan K.C

Intern at Geospatial Solution team at Munich Reinsurance
Deutschlandstipendium Scholarship 2024/25 and 2025/26

Lothstraße 62, 0005
80335 Munich
15.04.2001
Nationality: Nepali

Profile

Environmental engineering graduate specializing in Environmental Risk and Hazard, and Water Resources Management. Experienced in advanced GIS and Remote Sensing, Python Programming and Hazard Modeling.

Education

Apr 2024 – Present **Technical University of Munich, Germany**
Master of Science in Environmental Engineering
Specialization: Environmental Hazards and Risk, Water Resources Management
3rd semester **CGPA: 1.48**

Aug 2018 – May 2023 **Kathmandu University, Nepal**
Bachelor of Technology in Environmental Engineering
Key Subjects: Environmental (Water Resources, GIS)
CGPA: 1.3

July 2016 – April 2018 **Khwopa Secondary School, Bhaktapur**
Key subjects: Science Major

Honors and Awards

Deutschlandstipendium (2025-2026)

Awarded for academic excellence and commitment to social engagement.

Deutschlandstipendium (2024-2025)

Awarded for academic excellence and commitment to social engagement.

Experience

Sep 2025 – Present **Internship –Munich Re , Munich, Germany,**
Geospatial Solutions Team
Developing geospatial solutions for climate and hazard risk modeling, utilizing advanced Remote Sensing technology. Processing large-scale satellite products for disaster models and integration into risk analysis pipelines.
Building APIs to operationalize GIS and remote sensing algorithms for production-level risk assessment platforms.

Nov 2024 – Sep 2025 **Student Research Assistant, Technical University of Munich (TUM), Munich Germany**
Chair of Hydrology and River Basin Management
Conducting flood risk analysis using remote sensing, modeling, and advanced data techniques to support sustainable water and hazard management.

Tutor for graduate-level courses:
- Flood Risk and Flood Management
- Urban Hydrology and Urban Flood Modeling

June 2023 – April 2024

Environmental Engineer, *Research Institute for Sustainable Environment Pvt.Ltd.*

Engaged in Environmental impact assessments, Environmental modeling, GIS, and Remote sensing application in sustainable development.

April 2020 – March 2024

Freelancer GIS analyst, Upwork

Provided geospatial analysis for global clients, including suitability modeling, remote sensing data processing, and spatial data analysis. Work included developing an evapotranspiration (ET) model in Google Earth Engine (GEE) by integrating satellite-derived inputs (e.g., LST, soil moisture, LULC, Vegetation Indices, Surface Albedo, and net radiation flux etc.)

Projects

March 2025-Sep 2025

Improvement of Areal Precipitation for Rainfall-Runoff Modeling using Modified Topographic Information (Grade: 1.0)

Conducted as Part of a Master's Level Study Project in Environmental Engineering

Developed and implemented a novel directional DEM smoothing and stratified kriging framework, reducing precipitation estimation errors by 45.4% and enhancing the elevation-precipitation correlation for hydrological applications.

Quantified the impact of refined precipitation data on a calibrated HBV model, evaluating its utility for flood forecasting and analyzing critical interactions between input data and model structure.

Oct 2024- Feb 2025

Flood Risk Reduction in Roßhaupten, Ostallgäu : Study on Culvert Design and Runoff Mitigation Strategies to prevent flooding.

Conducted as Part of University Collaboration with Rosshaupten Municipality

Utilized high-resolution geospatial parameters (topography, land cover) in GIS and hydrological models for micro-level flood risk assessment. The project involved modeling the area's hydrological response, analyzing various runoff mitigation strategies (e.g., culvert design, retention basins), and applying analytical skills to spatial data to develop and deliver practical engineering recommendations to municipal partners.

July 2024- Sep 2024

Computer Programming in Hydrology - Incorporating radiation into the snowmelt runoff model.

Developed a physically-based snowmelt runoff model for the Inn River Basin, Bavaria, by integrating satellite-derived radiation data to improve model accuracy. This project required use of Google Earth Engine (GEE) for efficient remote sensing data retrieval and Python programming for model development, calibration, and hydrological result visualization.

Dec 2022 – May 2023

Study of the effects of climate change in the glaciers of Nepal: A case study of Marshyangdi river basin, Nepal.

This study investigated long-term glacier dynamics (1990 to 2020) in the Marshyangdi river basin, Nepal, to assess climate change effects. The work required intensive processing of multi-temporal satellite imagery (Landsat 4/5/8 and Sentinel-2) and the application of Machine Learning algorithms to accurately map and differentiate between clean and debris-covered glacier ice across the region.

August 2020

Post-Disaster Land Use/Land Cover (LULC) Change Detection
Conducted a study using Remote Sensing satellite imagery to analyze and quantify LULC change patterns in the Panauti and Banepa Valley following the major 2015 earthquake. The analysis focused on Change Detection and incorporated essential ground validation techniques to verify the accuracy of the derived geospatial data and results.

June 2019

Design and fabrication of lab scale anaerobic digestor.
Semester Project to design and fabricate an Anaerobic Digestor to digest the food waste from Kathmandu University Canteen.

Mentorships

March 2025

Empowering Resilience: GIS and Remote Sensing for Disaster Risk Reduction in Nepal.
Delivered an online session for undergraduate students on using GIS and remote sensing to support disaster risk reduction efforts.

April 2023

ArcGIS instructor for Undergraduate students.
Forum for Environmental Conservation and Management (FECAM)
Kathmandu University, Dhulikhel

Mar 2023

Workshop on Application of Satellite Generated Atmospheric and Climatic Dataset
University of Salzburg, ERASMUS+, and Kathmandu University
Address: Dhulikhel, Nepal

Volunteering

March 2025

Trainer – Online Awareness Session on Plastic Waste Management
Regen Nepal Pvt. Ltd., Nepal
Volunteered to deliver a capacity-building session for students, homemakers, and small business owners in Nepal. Focused on practical strategies for plastic segregation, recycling, and reuse to promote sustainable behavior and community-driven waste management solutions.

July 2023

Restoring Lives: Volunteer Efforts in Flood Relief and Resettlement in Sindupalchok, Nepal
Role: Engineered Solutions as a Junior Environmental Engineer

Sep 2022

Plantation campaign in Dhulikhel Municipality under British Council's Road to COP26 campaign.
[Dhulikhel]

Mar 2022

Facilitator for symposium "Sustainable Cities in Nepal, from Perception to Reality"
[Kathmandu University, Dhulikhel]

Key Skills

- ArcGIS/QGIS
- Python
- Google Earth Engine
- Remote Sensing
- Environmental Modeling
- HEC-RAS and HEC-HMS
- Data Analysis and Statistics
- API Development